



Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : III Semester B.E CSE B
Course Code & Title : CS3352 & Foundations of Data Science
Name of the Faculty member : Mrs. P. Sabeena Burvin, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit II / Graphs for Quantitative and Qualitative Data
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO 1,2,4,10
- **Activity Chosen:** Think-Pair-Share
- **Date of Implementation:** 21-08-2025

- **Justification:**

The **Think–Pair–Share** strategy encourages active participation and collaborative learning. In this session, students explored how different types of graphs are used to represent **quantitative** (numerical) and **qualitative** (categorical) data. The goal was to understand which visualizations are most effective for different data types and how graphical choices influence interpretation. Active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).

- **T (Think):** Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question.
 - **P (Pair):** Teacher asks another question, related to the previous one that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task.
 - **S (Share):** Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points.
- **Time Allotted for the Activity:** 30 Minutes
 - **Details of the Implementation:**

Objectives

- To differentiate between quantitative and qualitative data.
- To identify appropriate graphs for each data type.
- To interpret and compare data using visual representations.
- To collaborate and share ideas on effective data visualization practices.

Think Phase (Individual Work)

Each student individually reflected on the following questions:

1. What are some examples of quantitative and qualitative data?
2. Which graphs are suitable for displaying each type?
3. How can graphs mislead or distort information if used improperly?

Individual Notes:

- Quantitative data → numerical, can be measured (e.g., income, temperature, test scores).
 - Suitable graphs: Histograms, Box plots, Scatter plots, Line graphs.
- Qualitative data → descriptive, categorical (e.g., gender, color, brand).
 - Suitable graphs: Bar charts, Pie charts, Pareto charts.

Pair Phase (Collaborative Discussion)

Students formed pairs to discuss their thoughts and findings.

Discussion Points:

- Compared examples and justified graph selections.
- Debated the effectiveness of pie charts versus bar charts for qualitative data.
- Explored the role of color, labeling, and clarity in communicating data.
- Identified real-world scenarios (e.g., surveys, experiments, sales reports) where certain graphs are more meaningful.

Key Observations:

- Bar charts provide clearer category comparisons than pie charts when there are many categories.
- Histograms effectively show frequency distribution for quantitative data.
- Scatter plots are valuable for showing relationships or correlations between two variables.
- Simplicity and accuracy are critical for effective data communication.

Share Phase (Class Discussion)

Pairs presented their insights to the class. The instructor facilitated a group discussion to consolidate understanding.

Summary of Shared Ideas:

- Graph choice depends on data type and analysis purpose.
- Both data types can be combined in dashboards for richer insights.
- Misleading graphs can be avoided by ensuring proportional scaling and appropriate labeling.
- Visualization tools like Excel, Tableau, and Python (Matplotlib, Seaborn) make creating and customizing graphs easier.

Learning Outcomes:

After completing the Think–Pair–Share activity, students were able to:

- Classify data as qualitative or quantitative.
- Choose suitable graphs for data representation.
- Interpret and critique graphs effectively.
- Collaborate and communicate statistical ideas visually.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1
CO1	3	3	3	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12
	2	2	3	3	3	2

Justification for correlation	Apply basic Knowledge and fundamentals in describing data with graphs	Identify the need of describing data with graphs	Able to describe graphs for both qualitative and quantitative data	Functional individually in describing data with suitable graphs	Communicate/ share the ideas with other students in written and oral presentation	Ability to identify which graph is suitable one for real time data's through self-learning
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• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Working as a pair, they can share their knowledge with them and come up with better understanding about the topic.
- Different types of problem-based questions are discussed in the class room, so students come to know how to solve and calculate the averages and describe the data using formulas.

❖ **Benefit of the practice:**

- Students are actively engaged and students learn from each other.
- Makes class interactive and builds a friendly, yet academic atmosphere.
- Includes all the students in the teaching-learning process.

❖ **Challenges faced in implementation:**

- Students felt very difficult in this activity and could not complete in time whenever new real time system asked in the question.
- Students are hesitating to speak in front of all.

References:

- ❖ <https://www.readingrockets.org/strategies/think-pair-share>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2023-2024/IP/2.KVS_23_24_CS3352_TPS.pdf
- ❖ <https://serc.carleton.edu/introgeo/interactive/tpshare.html>